

# From Control to Command: A Bimanual Mimetic Gesture Interface for Drone Swarm Maneuvering in Distracted Environments

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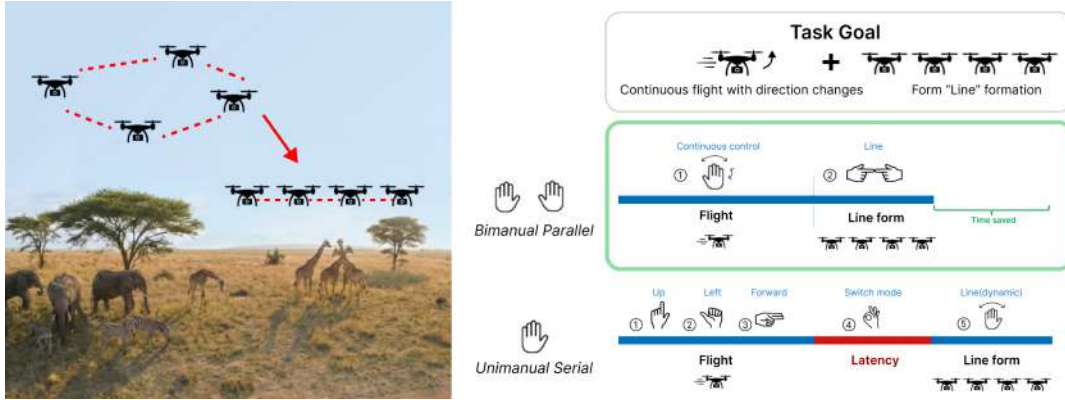


Figure 1: Typical operation flow diagram of drone swarm.

## Abstract

To address high cognitive load and latency in drone swarm control, we propose BiMime, an asymmetric bimanual system based on mimetic interaction. While traditional unimanual interfaces force inefficient serial switching between formation and flight, BiMime enables parallel execution by assigning discrete formation management to the left hand ("Commander") and continuous motion to the right hand ("Pilot"). By mapping hand spatial configurations directly to swarm geometries through intuitive metaphors, the system significantly reduces cognitive complexity. A preliminary pilot study suggests that BiMime shows promise in reducing task completion time, error rates, and workload (NASA-TLX) while achieving high usability (SUS), validating its efficacy in demanding tactical environments.

\*These authors contributed equally to this research.

## CCS Concepts

• Human-centered computing → Gestural input; User interface design.

## Keywords

Touch/Haptic/Pointing/Gesture, Interaction Design, Drone Swarm Control, Bimanual Interaction, Mimetic Interaction, Cognitive Load

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## 1 Introduction

Gesture-based interaction is critical for intuitive swarm orchestration. However, unimanual interfaces are subject to a "single-channel bottleneck" [17, 20], which compels operators to issue navigation and formation commands sequentially in dynamic missions such as jungle anti-poaching [5]. This serialization introduces tactical



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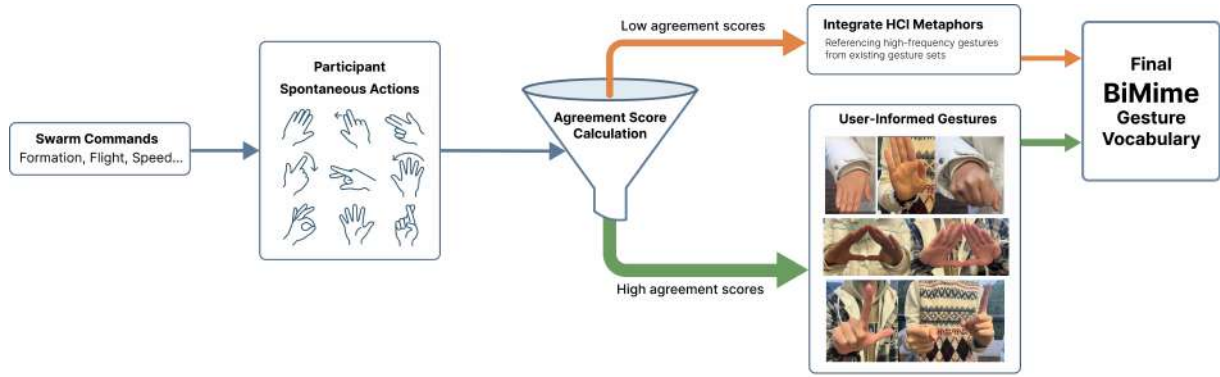


Figure 2: Design process.

latency and often exceeds human attentional limits, particularly in fast-paced, high-stakes operational scenarios.

To resolve this, we propose BiMime, an asymmetric framework grounded in kinematic chain and mimetic isomorphism principles. By delegating discrete states to a "Commander" (left hand) and continuous motion to a "Pilot" (right hand) [9, 11], BiMime enables concurrent control while mapping hand configurations directly to swarm geometries. This isomorphic mapping reduces "cognitive dimensionality" [16] by allowing operators to intuitively "shape" the swarm through hand gestures. Pilot evaluations demonstrate substantial improvements in reaction time and accuracy compared to traditional unimanual systems.

The primary contributions of this work include: the BiMime architecture for concurrent swarm formation and flight control; the Commander-Pilot paradigm, which eliminates the serial bottlenecks of unimanual systems; an isomorphic gesture vocabulary based on intuitive physical metaphors; and empirical validation via a pilot study showing significantly reduced reaction time and error rates in tactical maneuvers.

## 2 Related Work

### 2.1 Limitations of Current Modalities

Existing interaction modalities face bandwidth and robustness limitations in swarm control. Physical controllers provide precision but suffer from hierarchical menu logic unsuitable for dynamic formations [12]. Alternatives like Natural Language [4] and Eye Tracking [18] free the hands but lack the fidelity for complex trajectories and are sensitive to noise or lighting. Vision-based systems using skeletal tracking (e.g., MediaPipe [10]) enable real-time gesture control for drones [13] and swarms [1]. However, they are predominantly unimanual, creating a serial bottleneck: operators must stop flying to change formations, leading to latency and disrupted fluidity.

### 2.2 Theoretical Foundation for BiMime

To overcome this, our work is grounded in a dual-theoretical foundation:

- **Asymmetric Bimanual Collaboration:** Based on the Kinematic Chain Model [7], we assign the non-dominant hand as

the formation anchor and the dominant hand for fine flight control, enabling parallel command execution.

- **Mimetic Intuition:** Moving beyond abstract "Legacy Bias" gestures [6], we employ physical isomorphic mapping. Bimanual postures directly correspond to swarm structures, aligning with the user's mental model for intuitive, "what-you-see-is-what-you-get" formation control.

## 3 Gesture Set Design

### 3.1 Design Methodology and Elicitation

To ensure alignment with user mental models, the development process began with a guessability study [15], as illustrated in Figure 2. The elicitation phase involved three undergraduate students (2 males, 1 female;  $M_{age} = 21.3$ ) from the department of Industrial Design. All were right-handed novices with no prior experience in drone operation, ensuring that the elicited gestures represented intuitive, non-expert mental models. During the elicitation phase, participants were prompted with verbal commands to perform intuitive hand movements, revealing a high degree of consensus across four core functional categories: Global Movement, Circular Formation, Speed Change, and Hovering. These high-agreement gestures were adopted as user-elicited anchors. Furthermore, to address abstract commands and ensure system scalability, our design integrated established **HCI Metaphors**—bridging the gap between physical intuition and digital control logic. This systematic approach culminated in BiMime's final gesture vocabulary (Figure 3), characterized by three meta-strategies: Physical Isomorphism, Instrumental Interaction, and Asymmetric Embodiment.

*Asymmetric Bimanual Collaboration: The Commander-Pilot Paradigm.* Based on the Kinematic Chain Model [7], BiMime assigns distinct functional roles to each hand, mimicking the asymmetric division of labor in professional aviation to decouple complex multi-agent tasks.

- **The Left Hand (Commander) & Physical Isomorphism:** Responsible for discrete state management. We employ Physical Isomorphism, where hand spatial structures directly simulate the swarm's geometry—a 3D extension of the "What You See Is What You Get" (WYSIWYG) principle. For example,

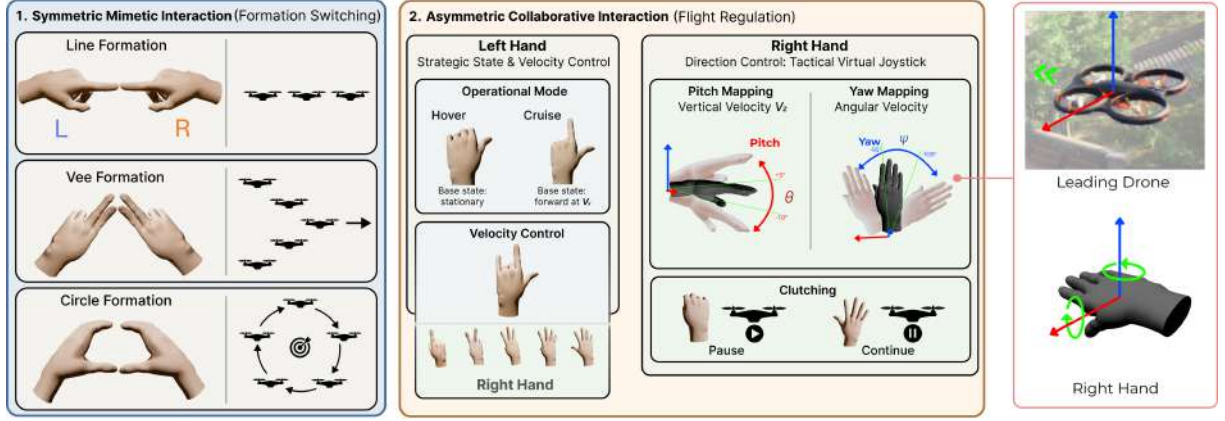


Figure 3: Gesture Vocabulary.

positioning index fingers into a line triggers a Line Formation, while a V-shape configuration initiates Vee Formation. This mimetic mapping eliminates the cognitive translation cost between abstract symbols and physical outcomes.

- **The Right Hand (Pilot) & Instrumental Interaction:** Manages high-frequency, continuous maneuvers. Following the Instrumental Interaction framework [2], the hand acts as a "virtual tool" rather than a direct manipulator, providing a layer of abstraction necessary for stable flight.
- **Coordinated Command Stacking & Semiotic Metaphors:** This decoupling allows parallel execution: formation changes (left) occur concurrently with flight (right). For abstract parameters, we integrate Semiotic Metaphors based on social consensus. For Velocity Control, the left hand initiates a mode-set while the right hand uses a Counting Metaphor (finger counts 1–5) to specify values, ensuring safety through coordinated intent.

*Right-hand Continuous Control: Virtual Joystick Metaphor.* Unlike traditional direct-position mapping, the right hand operates through a Virtual Joystick Metaphor using a rate-based control strategy. This reduces physiological strain and ensures task fluidity.

- **Rate-based Mapping:** Hand Euler angles are mapped to velocity components ( $\dot{x}$ ,  $\dot{y}$ ,  $\dot{z}$ ) rather than absolute coordinates [13].
  - **Pitch Mapping:** Hand pitch ( $\theta_{\text{hand}}$ ) controls vertical velocity ( $V_z$ ); extension (tilting up) causes ascent, while flexion (pressing down) causes descent.
  - **Yaw Mapping:** Hand yaw ( $\psi_{\text{hand}}$ ) determines the angular velocity ( $\omega_z$ ), enabling intuitive rotation of the swarm heading.
- **Safety & The Clutching Metaphor:** To resolve the "Midas Touch" problem in distracted environments, we implemented a central **Dead Zone** ( $-10^\circ < \theta_{\text{hand}} < +5^\circ$ ) to filter tremors. Furthermore, we introduce a Clutching Metaphor: toggled by making a fist, it allows the operator to "disengage" the control link to rest or reposition their arm without disrupting the swarm's stability—akin to a mechanical clutch in a vehicle.

### 3.2 Implementation and System Architecture

To prevent resource contention between visual processing and 3D simulation, BiMime employs a distributed architecture that ensures low-latency responsiveness.



Figure 4: The BiMime system architecture



Figure 5: Simulation scenario of the BiMime system.

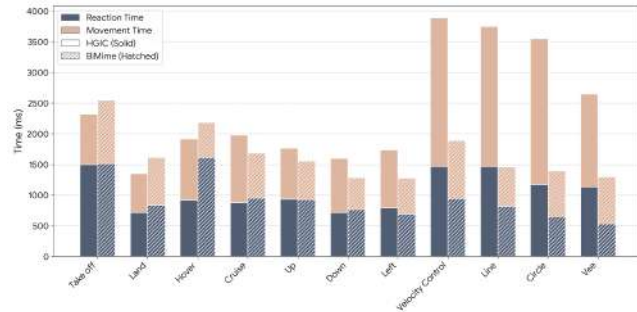
A macOS host serves as the perception module, utilizing the Google MediaPipe framework [10] to extract 21 3D hand landmarks at a stable 30 fps via a webcam. This spatial data is continuously streamed via the low-latency OSC/UDP protocol to a Windows simulation host. On the Windows machine, a Python bridge acts as the control logic center, translating the incoming coordinate streams into 6-DOF flight commands conditioned on the clutch state. These commands are executed within Unreal Engine 4 using the AirSim platform [14] to maneuver the UAV swarm (Figure 5). The system source code is available at: [https://github.com/PppPbb/BiMime\\_DroneSwarmGestureControl/tree/UE-Airsim\\_project](https://github.com/PppPbb/BiMime_DroneSwarmGestureControl/tree/UE-Airsim_project).

## 4 Evaluation

### 4.1 User Study

We conducted a within-subjects pilot study comparing BiMime with the baseline HGIC [8]. To ensure consistent task delivery and isolate cognitive evaluation from confounding technical variables, a video-based Wizard of Oz (WoZ) framework was utilized. In this setup, participants were presented with pre-recorded, high-fidelity tactical maneuvers of a drone swarm—such as the sudden appearance of obstacles. Although the swarm’s movement was predetermined and decoupled from real-time inputs, this approach ensured identical experimental conditions for all subjects, allowing us to focus exclusively on how intuitively participants could map their gestures to the swarm’s actions.

A cohort of six participants ( $N = 6$ ,  $M_{age} = 20.5$ ,  $SD = 2.17$ , 4 females, 2 males) was recruited for this evaluation. Among them, four were right-handed and two were ambidextrous; their prior drone experience ranged from never ( $n = 3$ ) to novice ( $n = 2$ ) or occasional ( $n = 1$ ). To mitigate order effects, an AB/BA counterbalancing scheme was implemented, with participants randomly assigned to start with either BiMime or the HGIC system. In this pilot, each participant performed a single, comprehensive trial for each system category. Although categorized as novices in swarm commanding, all participants were screened for foundational competence in spatial-gestural interaction to ensure high-quality user feedback, allowing for an in-depth assessment of the system’s intuitive mapping before future large-scale deployment.



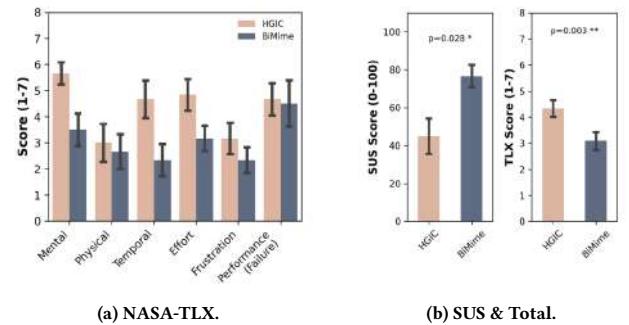
**Figure 6: Experimental results of Reaction Time (RT) and Movement Time (MT).**

### 4.2 Results

**4.2.1 Operational Efficiency: Task Completion Time (RT & MT).** BiMime significantly outperformed HGIC in MT (See Figure 6), primarily because HGIC requires frequent serial mode-switching. Although RT was similar for basic gestures, HGIC exhibited "Online Planning"—participants hesitated during mode-switch execution, shifting cognitive load from the planning phase to the execution phase. Furthermore, HGIC’s multi-step logic caused "Command Stacking" failures, where rapid command sequences overwhelmed participants, leading to frequent missed responses. The single-step BiMime interaction successfully eliminated this temporal bottleneck.

**4.2.2 Interaction Accuracy and Error Pattern Analysis.** Error analysis revealed a fundamental shift in user mental models. HGIC suffered from high Cognitive Errors (e.g., confusing "Line" and "Vee"), proving that abstract gestures lack the robustness required for high-pressure swarm command. Conversely, BiMime errors were limited to "Behavioral Over-action" (e.g., instinctive movement of the non-dominant hand), which did not impede system recognition. This shows that BiMime’s mimetic logic was successfully internalized, requiring minimal conscious recall.

**4.2.3 Subjective Workload and Usability.** NASA-TLX results indicate that BiMime significantly reduces the operator’s cognitive burden compared to HGIC. Specifically, BiMime achieved significantly lower scores in Mental Demand ( $M_{BiMime} = 3.50$  vs.  $M_{HGIC} = 5.67$ ,  $p = 0.041$ ), Temporal Demand ( $p = 0.009$ ), Effort ( $p = 0.011$ ) and Frustration ( $p = 0.004$ ). This efficiency comes from BiMime’s asymmetric bimanual logic, which optimizes resource distribution by delegating discrete parameter adjustments (e.g. formation switching) to the left and continuous spatial control to the right. This separation minimizes the internal resource competition inherent in the HGIC single-channel unimanual scheme, which forces the concurrent management of discrete states and continuous maneuvers. To complement the cognitive workload assessment, we employed the **Borg CR10 Scale** [3] to measure perceived physical exertion and task difficulty. The Borg CR10 is a category-ratio scale anchored by a semi-logarithmic increase in intensity (from 0, "Nothing at all," to 10, "Extremely strong/Maximal"), allowing participants to report physical strain with higher sensitivity to extreme levels than linear scales. In our study, BiMime yielded lower exertion scores, validating that the "Commander-Pilot" paradigm and the clutching mechanism effectively mitigate the physiological fatigue often associated with continuous mid-air gestural interaction.



**Figure 7: System evaluation results.**

Regarding overall usability, the SUS scores revealed a stark contrast: BiMime averaged 76.67 ("Good"), while HGIC scored significantly lower at 45.00 ("Fail") ( $t(5) = -3.06$ ,  $p = 0.028$ ). This disparity highlights the superiority of Mimetic Mapping over abstract referential gestures. Because BiMime’s gestures directly mirror the physical morphology of the swarm, they establish an intuitive mental model that minimizes the recall effort. Qualitative feedback confirmed that while HGIC reduces hardware reliance via its unimanual approach, the resulting operational complexity creates an unfavorable trade-off that hinders fluid swarm management.

### 4.3 Practical Significance in the Jungle Anti-Poaching Scenario

The lower RT of BiMime enables rapid response to emergent poaching indicators. The optimized MT ensures immediate obstacle-avoidance maneuvers. The low ER ensures mission safety and stealth. This efficient and robust interaction paradigm provides solid support for individual personnel commanding drone swarms in harsh natural environments.

## 5 Discussion

The clutching mechanism and dead-zone (detailed in Section 3) serve as our core countermeasures for robust deployment in distracted environments.

A critical challenge in dynamic, high-load contexts (e.g., field operators subject to frequent interruptions) is the "Midas Touch" problem, where always-on continuous tracking misinterprets resting postures or non-command gestures as flight inputs. By enforcing an active clutch, BiMime provides the essential "input safety" required when an operator's visual or cognitive attention is temporarily diverted away from the swarm.

While our pilot study provides promising directional evidence for BiMime, we acknowledge its methodological constraints. Although the limited scale of the evaluation constrains broad statistical generalizability, participants' ability to successfully maneuver a 6-DOF swarm with minimal training strongly highlights the interface's low barrier to entry and baseline intuitiveness. Furthermore, while our Wizard-of-Oz setup ensured internal validity without explicit external distractors, the built-in clutch design fundamentally prepares the interface for real-world unpredictability.

## 6 Conclusion and Future Work

BiMime effectively resolves the cognitive bottlenecks inherent in swarm control by employing an asymmetric bimanual paradigm that maps complex formations onto intuitive physical metaphors. This decoupling empowers operators to manage mission strategy and flight maneuvers in parallel. To validate these preliminary findings, future work entails a comprehensive summative evaluation with a diverse cohort—including expert drone operators—testing the system against explicit secondary distractors in physical environments. Future iterations will integrate IMU-equipped tactile gloves with vision tracking to mitigate arm fatigue and resolve visual occlusion issues. This hardware evolution—potentially incorporating surface electromyography (EMG) patches to provide physiological feedback of swarm status [19]—will support "micro-gestures," enabling high-precision control with minimal physical exertion. Simultaneously, we aim to achieve fluid human-algorithm coordination by introducing advanced flight logic that autonomously handles low-level stabilization and obstacle avoidance. The ultimate goal is a seamless interaction where the algorithm intelligently interprets intent, allowing the operator to maintain a direct connection to the swarm's real-time movement.

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